

Executive Summary & Central Conclusion

CONFIDENTIAL & PRE-CLINICAL

Integrated Multi-Omics Target Validation for In Vivo Type 2 Diabetes (T2D) Study

Framework: Xu et al. (2026)
Date: September 2026

1. Strategic Overview & Analytic Funnel

This study establishes a rigorous, end-to-end multi-omics target nomination and validation pipeline for Type 2 Diabetes (T2D). By integrating **human liver bulk transcriptomics (WGCNA)**, **KEGG pathway topology (KGML)**, **mouse single-cell RNA sequencing (liver & kidney atlases)**, **machine learning consensus modeling**, and **Mendelian Randomization (MR)**, we identify the primary mechanistic regulatory axis driving diabetic tissue stress and nominate prioritized targets for in vivo pharmacological validation.



2. The Central Biological Conclusion

Suppression of the AMPK Signaling Axis Output

The central biological finding across human clinical cohorts and rodent diabetic models is the **functional suppression of AMPK pathway output activity** under chronic hyperglycemic/insulin-resistant stress:

Suppressed AMPK Activity ⇒ $\begin{cases} \text{PPARGC1A (PGC-1}\alpha\text{)} \downarrow \\ \text{CCND1 (Cyclin D1)} \uparrow \end{cases}$

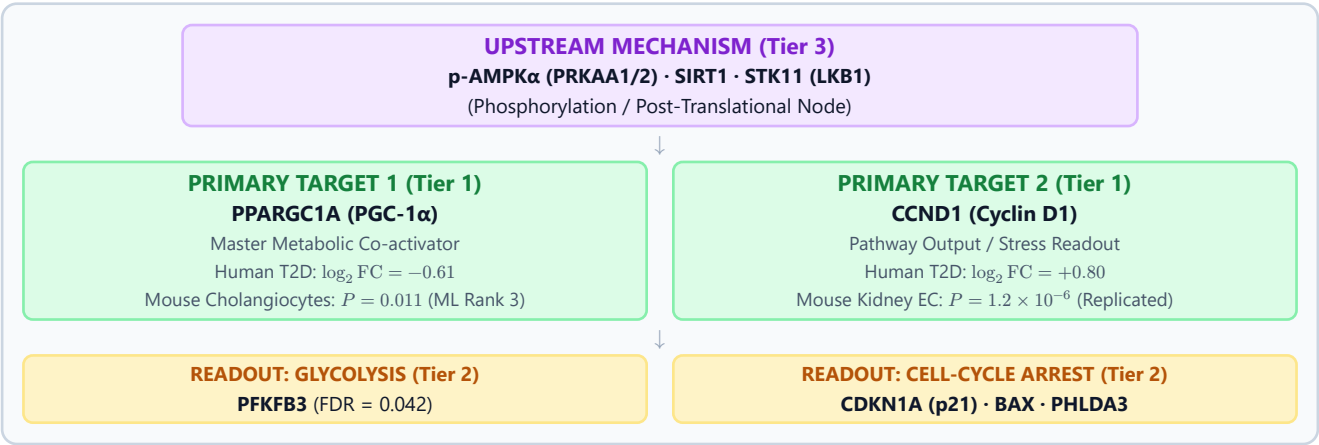
Loss of mitochondrial biogenesis & metabolic control
De-repressed cellular stress & aberrant proliferation signal

Key Mechanistic Insights:

- Post-Translational Regulatory Control:** The human KEGG network displays discordant edge dynamics: the AMPK kinase subunit (**PRKAA2**) is transcriptionally maintained while its activated downstream effectors fall. AMPK activity is governed by **Thr172 phosphorylation** and cellular energy charge, not transcript abundance.
- Epithelial & Microvascular Compartmentalization:**
 - In **liver**, metabolic suppression (**Ppargc1a** downregulation) localizes specifically to epithelial **cholangiocytes** (per-mouse $P = 0.011$, detection dropping from 31.7% to 11.4%).
 - In **kidney**, stress/cell-cycle de-repression (**Ccnd1** upregulation) localizes to **microvascular endothelial cells** (pseudobulk $P = 1.2 \times 10^{-6}$).
- Cross-Species Concordance:** The directional shifts of both primary anchors (**PPARGC1A** down, **CCND1** up) replicate across human clinical cohorts and mouse diabetic models.

3. Core Target Tiers & Evidence Hierarchy

Targets are classified mechanically into prioritized tiers based on evidence convergence across all five analytical stages:



Target Classification Table

Target	Tier	Role & Mechanism	Key Supporting Evidence	Planned In Vivo Assay
PPARGC1A (PGC-1α)	Tier 1 (Primary)	Master transcriptional coactivator of mitochondrial biogenesis & respiration.	- Human liver: log₂ FC = −0.61, <i>P</i> = 6.0 × 10^{−4} - Mouse cholangiocytes: <i>P</i> = 0.011, cell BH <i>P</i> = 2.0 × 10^{−8} - ML Consensus: Rank 3 / 19	- RT-qPCR & Western blot (liver) - IF co-stain: PGC-1α × CK19
CCND1 (Cyclin D1)	Tier 1 (Primary)	De-repressed downstream output of AMPK → HuR/ELAVL1 axis.	- Human liver: log₂ FC = +0.80, <i>P</i> = 7.5 × 10^{−4} - Replicated in cohort GSE23343 (<i>P</i> = 0.016, AUC = 0.81) - Mouse kidney EC: <i>P</i> = 1.2 × 10^{−6}	- RT-qPCR & Western blot (kidney & liver) - IF co-stain: Cyclin D1 × CD31
CDKN1A (p21)	Tier 2 (Readout)	Cell-cycle arrest / senescence effector downstream of stress.	- Human liver DEG: log₂ FC = +0.96, FDR = 0.016 - Elevated in mouse cholangiocytes (FDR = 0.009) & kidney EC (FDR = 0.008)	- RT-qPCR & Western blot - IF co-staining with CD31 in kidney
PFKFB3	Tier 2 (Readout)	Inducible glycolytic regulator connected to AMPK.	- Human liver: FDR = 0.042 - Mouse liver LSEC (<i>P</i> = 0.049) & kidney LoH (BH <i>P</i> = 0.03)	RT-qPCR & Western blot (liver & kidney)
PRKAA1/2 (AMPKα)	Tier 3 (Master)	Central metabolic sensor and upstream master kinase.	Upstream driver on KEGG hsa04152; acts via Thr172 phosphorylation.	- Western blot for p-AMPKα (Thr172) / total AMPKα - Metformin control

4. Multi-Modal Evidence Matrix

Gene	Human Bulk	WGCNA	KEGG Map	Mouse scRNA	Replication	ML Rank	cis-MR	Tier
PPARGC1A	Down (<i>P</i> = 6.0 × 10 ^{−4})	Key Module	Direct (+1)	Down (Cholang. <i>P</i> = 0.011)	Concordant	Rank 3	Null (<i>P</i> = 0.16)	Tier 1
CCND1	Up (<i>P</i> = 7.5 × 10 ^{−4})	Key Module	2-Step (−1)	Up (Kidney EC <i>P</i> = 1.2 × 10 ^{−6})	Replicated	Rank 14	Null (<i>P</i> = 0.55)	Tier 1

Gene	Human Bulk	WGCNA	KEGG Map	Mouse scRNA	Replication	ML Rank	cis-MR	Tier
CDKN1A	Up (FDR = 0.016)	Key Module	p53 Map	Up (Cholang. & EC)	Concordant	—	Null	Tier 2
PFKFB3	Down (FDR = 0.042)	Turquoise	Direct (+1)	Down (LSEC $P = 0.049$)	Concordant	—	Null	Tier 2
IRS2	Down ($P = 0.011$)	Key Module	Input Node	Minor cells only	Concordant	Rank 6	Null	Tier 2
IGF1	Down ($P = 0.021$)	Key Module	Input Node	Sparse (Hepatocyte gap)	Concordant	Rank 8	Null	Tier 2b
PRKAA2	Up ($P = 0.017$)	Network Node	Master Kinase	Post-translational	Concordant	—	Null	Tier 3
SREBF1	Down ($P = 0.041$)	Network Node	Lipogenic	Inverted (STZ model)	—	—	Causal ($P = 9 \times 10^{-9}$)	Exploratory
VSNL1/LZTFL1	DE (FDR < 0.05)	Key Module	Off-Map	Not DE in mouse	Weak	Rank 1 & 2	Null	Non-Target

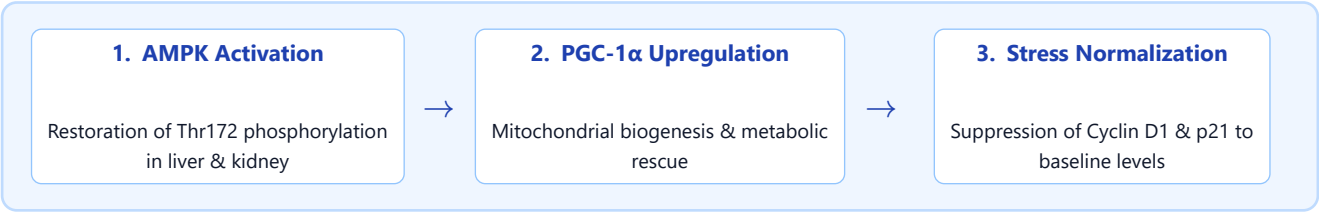
5. In Vivo Experimental Validation & Pharmacological Rescue Plan

The multi-omics analysis directly informs the experimental design, dosing groups, and molecular endpoints for the in vivo pre-clinical validation study in the Ob-T2D rodent model:

Experimental Groups (n = 8 per group)	Molecular & Phenotypic Endpoints
1. Normal Control: Standard diet + Vehicle (Saline). 2. Ob-T2D Disease Model: High-Fat Diet (HFD) + STZ. 3. Ob-T2D + GV1 Low Dose: Test formulation (Low). 4. Ob-T2D + GV1 Medium Dose: Test formulation (Med). 5. Ob-T2D + GV1 High Dose: Test formulation (High). 6. Ob-T2D + Metformin: Positive control (200 mg/kg).	• Primary Metabolic Efficacy: Oral Glucose Tolerance Test (OGTT), Fasting Blood Glucose, HbA1c, HOMA-IR. • Hepatic AMPK Activation: Western blot for p-AMPKα (Thr172) / total AMPKα. • Mitochondrial Function: PGC-1α (RT-qPCR & Western blot). • Cellular Stress / Proliferation: Cyclin D1 & p21 (qPCR & WB). • Immunofluorescence: PGC-1α × CK19 (liver cholangiocytes); Cyclin D1 × CD31 (kidney microvascular endothelium).

Hypothesized Therapeutic Mechanism of Action

Administration of the active herbal/natural formulation (**GV1**) is hypothesized to mirror the therapeutic action of **Metformin** through multi-target pathway restoration:



Document Provenance: This report synthesizes outputs from 24_target_scoring.R, 16_pathway_selection.R, 35_fig3_paper_style.R, 21_ML_classifier.R, and 19_mendelian_randomization.R. All supporting data tables, figures, and analysis scripts are cataloged in docs/target_selection.md and docs/pathway_selection.md.